

UCT512: CLOUD, MICROSERVICES AND APPLICATION

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2	1	2	3.5

Course Objectives: The course intends to introduce students to the fundamentals of developing application on Cloud, specifically public clouds such as AWS, AZURE and Google. Students would be able to appreciate:

1. How to design applications for Cloud
2. Develop applications using various services
3. Deploy applications on Cloud by using cloud native services

Cloud Fundamentals: Cloud Fundamentals, Models-IaaS/ PaaS / SaaS, NIST definition of Cloud, Cloud advantages deployment process. Public cloud Platform features-AWS, Azure, Google.

API Fundamentals Microservice fundamentals: Current system architecture, target system. Design approach. Design using spring boot project. examples. Sam Newman–Lectures on Micro services (references), API/Microservice design and deployment steps, Project use cases covering this

Devops fundamentals: Devops Tools and their usage in cloud application development, Docker and Containerization Process.

Python: Overview, Use cases for cloud application development.

Cloud Security and Monitoring Tools: Covering the cloud security and monitoring tools.

Tutorial Work: Design and developing solution steps using containers-containerization of application and deployment using Kubernetes, Projects use cases covering this.

Laboratory Work:

To select a cloud-based project and to design and implement applications on public cloud platforms such as AWS, AZURE and Google. To design use cases and implement cloud application using python.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Comprehend the basic concepts and architecture of Cloud computing.
2. Implement Cloud Services through AWS offerings and Restful web services.
3. Apply the knowledge of virtualization through different virtualization technologies.
4. Design use cases and implement Cloud-based project in python.

Text Books:

1. Buyya K, R., Broberg J. and Goscinski M. A., Cloud Computing: Principles and paradigms, MIT Press (2011) 4th ed. paradigms, MIT Press (2011) 4th ed.

- 2 Kai Hwang, Geoffrey Fox and Jack Dongarra, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Morgan Kaufmann (2012) 2nd ed.
- 3 Miller M., Cloud Computing, Que Publishing (2008) 1st ed.
- 4 Puttini R. and Mahmood Z., Cloud Computing: Concepts, Technology & Architecture, Service Tech press (2013) 1st ed.

Reference Books:

1. Velte A., Velte T., and Elsenpeter R., Cloud Computing: A practical Approach, Tata McGraw Hill (2009) 1st ed.
2. Hurwitz J., Bllor R., Kaufman M. and Halper F., Cloud Computing for dummies (2009) 1st ed.